Day 103 | Unit 07 | Teacher plan

Understand and Use Percent

Explain and Critique Percent Reasoning

Standards: 6.RP.A.3c

Learning target: I can explain and Critique Percent Reasoning, and explain and check my reasoning.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Divide the part by the whole, then multiply by 100. Keep the same units in part and whole. The percent can be above 100 if the part is larger than the reference whole. Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

Day 103 | Unit 07 | Lesson

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01 / Learn

Divide the part by the whole, then multiply by 100. Keep the same units in part and whole. The percent can be above 100 if the part is larger than the reference whole. Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer.

Worked example 1

40 is what percent of 200?

1. Divide the part by the whole, then multiply by 100.

2. Keep the same units in part and whole.

3. 40 ÷ 200 × 100 = 20%

4. the reference whole is 200.

5. State the result in the form requested: 20%.

Answer: 20%

Worked example 2

125 is 125% of an unknown quantity w. Select all equations that find w.
A. w = 125 ÷ (125/100)
B. w = 125 + 125
C. (125/100)w = 125
D. w = 125 × (125/100)

1. Write part = decimal rate × whole.

2. The part is percent rate times whole.

3. Divide 125 by 5/4

4. w = 100.

5. Check by multiplication.

Answer: A, C - w = 125 ÷ (125/100); (125/100)w = 125

Day 103 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

40 is what percent of 80?

02 / Constructed response

20% of a number is 24. Find the number and check.

03 / Constructed response

20 is what percent of 100?

04 / Constructed response

12% of a number is 24/5. Find the number and check.

05 / Constructed response

24 is what percent of 40?

06 / Constructed response

60% of a number is 108. Find the number and check.

07 / Constructed response

84 is what percent of 140? State which quantity is the reference whole.

08 / Multiple select

16 is 40% of an unknown quantity w. Select all equations that find w.
A. w = 16 ÷ (40/100)
B. w = 16 × (40/100)
C. (40/100)w = 16
D. w = 16 + 40

09 / Numeric entry and explanation

A quantity changes from a reference whole of 100 to a compared amount of 40. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

10 / Constructed response

125% of a number is 150. Find the number and check.

Day 103 | Unit 07 | Answer key

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1. 50%

40 ÷ 80 × 100 = 50%; the reference whole is 80.

6.RP.A.3c | Constructed response

2. 120

Whole = 24 ÷ 1/5 = 120. Check 20% × 120 = 24.

6.RP.A.3c | Constructed response

3. 20%

20 ÷ 100 × 100 = 20%; the reference whole is 100.

6.RP.A.3c | Constructed response

4. 40

Whole = 24/5 ÷ 3/25 = 40. Check 12% × 40 = 24/5.

6.RP.A.3c | Constructed response

5. 60%

24 ÷ 40 × 100 = 60%; the reference whole is 40.

6.RP.A.3c | Constructed response

6. 180

Whole = 108 ÷ 3/5 = 180. Check 60% × 180 = 108.

6.RP.A.3c | Constructed response

7. 60%

84 ÷ 140 × 100 = 60%; the reference whole is 140.

6.RP.A.3c | Constructed response

8. A, C - w = 16 ÷ (40/100); (40/100)w = 16

The part is percent rate times whole. Divide 16 by 2/5; w = 40. Check by multiplication.

6.RP.A.3c | Multiple select

9. 40%; denominator 100

Percent = (40 ÷ 100) × 100 = 40%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

10. 120

Whole = 150 ÷ 5/4 = 120. Check 125% × 120 = 150.

6.RP.A.3c | Constructed response